

START ME UP!

Entrepreneurship and AI are booming... but what happens when the two meet?

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Hephaestus (a mechanical Greek mythology god), Mary Shelley's Frankenstein's Monster, Pygmalion's Galatea, and countless other examples are testament to the human fascination with AI. Mathematical calculators were first devised in Ancient China in the form of an Abacus. In 1642, Blaise Pascal invented a clever calculator made of gears. Charles Babbage designed the Analytic Engine in the 19th century, and it is widely considered to be one of the first designs for a general-purpose computer. Ada Lovelace, the first programmer in the world who wrote an algorithm for a machine to print Bernoulli numbers, believed that machines could compose complex pieces of music. Needless to say, machines, or the ideas of them, have fascinated us for all of known human history.

ORIGINS

Mathematicians like Gottfried Leibniz and Thomas Hobbes were extremely interested in the systemisation of thought and logic as early as the 17th century. The question of whether thought could be broken into mathematics was unanswered for almost 200 years before an Austrian of the name of Kurt Gödel gave us something called

the Incomplete Theorems. Its implications were twofold – one, that there is a definite limit to what can be achieved using logic, and two that within these limitations, the formal approach works remarkably well.

In his seminal paper *Computing Machinery and Intelligence*, Alan Turing proposed a different way to determine whether an entity is human or a machine. In the introductory section he considered the question “Can machines think?”. He wrote that such a question was all but redundant in the quest to differentiate the two. Instead, he proposed a replacement that he called ‘The Imitation Game’. It is now commonly known as the Turing Test. This was in 1950.

The Dartmouth Workshop was held in 1956. It was the birthplace of the field of the “AI” that we are familiar with nowadays. It was an eight week brainstorming session, with scientists discussed neural networks, natural programming language, theory of computation, and creativity.

Today, AI touches everyone's lives in some way, pretty much on a daily basis. The ads you see, the manipulation of big data, the fuzzy logic sensor that washes your clothes... almost every service, smartphone and mass produced product has varying levels of AI involved in the process, somewhere. However, if you thought it was already all pervasive, wait a few years, it's set to take over.

STARTUPS

With AI taking over every field, it's no surprise to anyone that startups working on AI are hot right now. The future is AI, and the future giant companies of the world (the Facebook, Apple and Google of tomorrow) might start off as a small AI startup today. That said, Google, Microsoft, IBM, Facebook, Apple, Uber, and even Shell are investing heavily in AI. Here's a video of some smart people talking about AI that you should watch – <http://digit.in/elonmskai> – including Elon Musk and Sam Harris.

But enough of that. Let's take a look at the most interesting AI startups that we feel are doing very interesting work.

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ECHOBX

Founded by two Cambridge graduates, this startup claims to have developed the world's first AI that truly understands the “meaning of content”. Echobox understands the constantly evolving landscape of social media so as to try and maximise every brand's reach. Even casual users of Facebook and Twitter know that there is a number of different variables that allow one's post to have the maximum possible reach. Echobox takes into account all of these variables so as to direct different content to the required target audience.

It is a sign of the times that social media accounts have gone from being